Lecture 25: Lines and Planes in Space

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Write the vector, parametric, and cartesian equations of a line in three-dimensional space.
- ☐ Find the distance from a point to a line in space.
- ☐ Write the vector and component equations of a plane in three-dimensional space.
- ☐ Determine the line of intersection between two planes.
- ☐ Determine the line of intersection between a line and a plane.
- ☐ Find the distance from a point to a plane.
- ☐ Calculate the angle between two planes.

Vector Equation for a Line

Suppose that L is a line in space passing through a point $P_0(x_o, y_0, z_0)$ and parallel to a vector $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$.

Let P be a point on L, then $\overrightarrow{P_0P}$ is parallel to $\boldsymbol{v}$.

Vector Equation for a Line

A vector equation for the line L through $P_0(x_0, y_0, z_0)$ parallel to a nonzero vector v is

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + t\boldsymbol{v}, \quad -\infty < t < \infty,$$

where $\boldsymbol{r}$ is the position vector of a point $P(x, y, z)$ on L and $\boldsymbol{r}_0$ is the position vector of $P_0(x_0, y_0, z_0)$.

Parametric Equations for a Line

Parametric Equations for a Line

The standard parametrization of the line through $P_0(x_0, y_0, z_0)$ parallel to a nonzero vector $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$ is

$$x = x_0 + tv_1, \quad y = y_0 + tv_2, \quad z = z_0 + tv_3, \quad -\infty < t < \infty.$$

Example: Find parametric equations for the line through $(-1, 0, 4)$ parallel to $\boldsymbol{v} = 2\boldsymbol{i} + 4\boldsymbol{j} - 2\boldsymbol{k}$.

Parametric Equations for a Line

Example: Find parametric equations for the line through $P(-3, 2, -3)$ and $Q(1, -1, 4)$.

Parametric Equations for a Line Segment

To parametrize a line segment joining two points, we first parametrize the line through the points.

Then, we find the t-value at the endpoints and restrict t to fall in the closed interval bounded by these values.

Example: Parametrize the line segment joining the points $P(-3, 2, -3)$ and $Q(1, -1, 4)$.

Cartesian Equation

Cartesian Equation for a Line

The Cartesian equation of the line through $P_0(x_0, y_0, z_0)$ parallel to a nonzero vector $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$ is

$$\frac{x - x_0}{v_1} = \frac{y - y_0}{v_2} = \frac{z - z_0}{v_3} \quad (= t).$$

If v_1, v_2 or v_3 is 0, then x, y or z will, respectively, be a constant.

Example: Find the Cartesian equation for the line through $(2, -3, 1)$ parallel to $\boldsymbol{v} = 3\boldsymbol{i} + 6\boldsymbol{k}$.

Distance from a Point to a Line

Distance from a point S to a Line Through P Parallel to $\boldsymbol{v}$

$$d = \frac{|\overrightarrow{PS} \times \boldsymbol{v}|}{|\boldsymbol{v}|}$$

Distance from a Point to a Line

Example: Find the distance from the point $S(1, 1, 5)$ to the line

$$L: \quad x = 1 + t, \quad y = 3 - t, \quad z = 2t.$$

Equation for a Plane in Space

Equation for a Plane

The plane through $P_0(x_0, y_0, z_0)$ normal to a nonzero vector $\boldsymbol{n} = A\boldsymbol{i} + B\boldsymbol{j} + C\boldsymbol{k}$ has

$$\text{Vector equation} : \boldsymbol{n} \cdot \overrightarrow{P_0P} = 0$$

$$\text{Component equation} : A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

$$\text{Component equation simplified} : Ax + By + Cz = D, \text{ where}$$

$$D = Ax_0 + By_0 + Cz_0$$

Equation for a Plane in Space

Example: Find an equation for the plane through $P_0(-3, 0, 7)$ perpendicular to $\boldsymbol{n} = 5\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$.

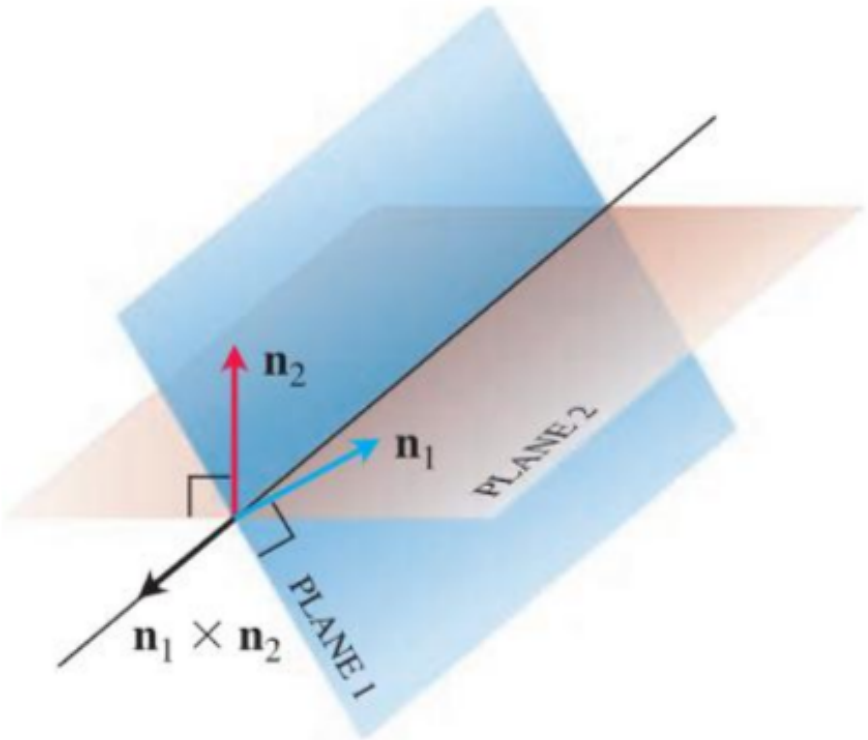
Equation for a Plane in Space

Example: Find an equation for the plane through $A(0, 0, 1)$, $B(2, 0, 0)$, and $C(0, 3, 0)$.

Intersection of Two Planes

Just as lines are parallel if and only if they have the same direction, two planes are parallel if and only if their normals are parallel.

Two planes that are not parallel intersect in a line.



Intersection of Two Planes

Example: Find the equation of the line of intersection of the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Intersection of a Line and a Plane

Example: Find the point where line

$$x = \frac{8}{3} + 2t, \quad y = -2t, \quad z = 1 + t$$

intersects the plane $3x + 2y + 6z = 6$.

Intersection of a Line and a Plane

Example: Determine where the line

$$x = -1 + t, \quad y = 2 + 5t, \quad z = 5 + t$$

intersects the plane $x - y + 4z = 5$.

Intersection of a Line and a Plane

Example: Determine where the line

$$x = -1 + t, \quad y = 2 + 5t, \quad z = 5 + t$$

intersects the plane $x - y + 4z = 17$.

Distance from a Point to a Plane

Distance from a Point S to a Plane Through a Point P with a Normal $\boldsymbol{n}$

$$d = \left| \overrightarrow{PS} \cdot \frac{\boldsymbol{n}}{|\boldsymbol{n}|} \right|$$

Distance from a Point to a Plane

Example: Find the distance from $S(1, 1, 3)$ to the plane $3x + 2y + 6z = 6$.

Distance from a Point to a Plane

Example: Find the distance from $S(2, -1, 3)$ to the plane $3x - 26y + 7z = -90$.

Angles Between Planes

The angle between two planes is the same as the angle between their normal vectors.

Example: Find the angle between the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.